

Problem Set 2 (QBCS 5002: Introduction to Quantitative Biology)

Due: October 13, 2026, 12:00 p.m.

The problem set is to be completed by teams of two. We strongly encourage teams pairing a student from a biology, medicine or chemistry background with a student from a mathematics, physics or engineering background. Each team will submit one joint solution, and both partners will receive the same grade.

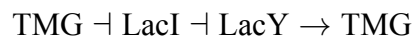
Your solutions should show the main algebraic steps. For the coding problems, you may use Python, Mathematica, MATLAB, C/C++, or another language approved by the instructor. Submit readable source code with comments identifying the key mathematical and numerical steps.

The complete submission should include a PDF report with commented source code in a separate file or as appendix. Cite the paper and external libraries or code resources used.

Problem 1-3 is based on the paper *Multistability in the lactose utilization network of Escherichia coli* (2004) and its Supplementary Information, which gives a detailed derivation of the mathematical model and explains how to estimate the model parameters from experiment.

Problem 1. Write code to visualize the rate plot and bifurcation diagram

In the lac operon system, the uptake of TMG induces the synthesis of LacY, which in turn promotes further TMG uptake. The resulting positive feedback loop is:



Let x denote intracellular TMG concentration and y denote LacY concentration. The Supplementary Information models the active fraction of LacI as

$$\frac{R}{R_T} = \frac{1}{1 + x^n} \quad (\text{S1})$$

where n is the Hill coefficient. LacY production and TMG transport are described by

$$\tau_y \frac{dy}{dt} = \frac{\alpha}{1 + R/R_0} - y \quad (\text{S2})$$

$$\tau_x \frac{dx}{dt} = \beta y - x \quad (\text{S3})$$

Define the repression factor

$$\rho = 1 + \frac{R_T}{R_0}$$

Note: In this problem set, we treat n as an arbitrary positive number in the range $1.5 \sim 4$, rather than adopting the approximation $n \approx 2$ as in the Supplementary Information. For the required numerical rate plots, bifurcation diagrams, and animations in Tasks 2-5, you may use $n = 2$ as a baseline, and explore how the results change for $1.5 \leq n \leq 4$.

Tasks

1. Derive the one-dimensional production-removal equation for a Hill coefficient of n . State the explicit expressions for the production and removal function $P(y)$, $R(y)$, and net rate $F(y) = P(y) - R(y)$, and derive the fixed-point equation for y .
2. Write code to plot the production and removal curve. Mark stable and unstable fixed points, and include a phase line indicating the direction of motion of y between fixed points.
3. Explore the roles of α , β , and ρ . Explain how each parameter changes the production curve.
4. Create a short animation or video to illustrate the saddle-node collisions as β changes slowly through the bistable regime.
5. Construct a bifurcation diagram with a proper control parameter. Explain how the bifurcation diagram predicts hysteresis.

Problem 2. Parameter estimation

The paper's single-cell experiment varies two external inputs, the extracellular concentrations of glucose (G) and TMG (T), and measures the resulting levels of two fluorescent reporters: green fluorescence reports lac promoter activity and serves as a proxy for lacY expression y ; red fluorescence reports CRP-cAMP activity through a gat promoter reporter.

The two fluorescence values under bistable conditions do not by themselves provide enough information to uniquely specify the underlying parameters α , β and ρ . However, by also applying the saddle-node condition at the switching thresholds, we are able to determine their complete functional dependence on G and T .

In experiment, the authors prepare cells with known history: for initially uninduced cells, test whether they switch ON; for initially induced cells, test whether they switch OFF. At each glucose concentration G , this gives two TMG thresholds T_{ON} and T_{OFF} (as shown in fig 2b and 2c). At each switching threshold, the single-cell GFP histogram gives the expression levels of the low and high expression branches: y_{low} and y_{high} (as shown in fig 4a). Near a saddle-node boundary, these two observed fluorescence branches correspond to the two distinct roots of y . We therefore can estimate the parameters at the threshold. Schematically:

$$(T_{\text{switch}}, y_{\text{low}}, y_{\text{high}}) \xrightarrow{\text{saddle-node fit}} \alpha, \beta, \rho$$

Tasks

1. At a switching threshold, the production and removal curves are tangent. If $y = a$ denotes the colliding fixed point, the saddle-node conditions are

$$F(a) = 0 \quad F'(a) = 0.$$

Derive a parametric description of the saddle-node boundary.

2. Derive the critical point at which the two saddle-node boundaries merge, and obtain expressions for $\rho_c(n)$ and $(\alpha\beta)_c(n)$. Verify that your result reduces to the paper's result when $n = 2$.
3. For the three model parameters α , β , and ρ , state how changing the glucose and TMG concentrations change the parameter values.
4. Explain how the model parameters can be determined from measurements near the switching thresholds. Write code to perform the following parameter estimation analysis.

Since the original experimental data are not available, use image digitization to obtain approximate data from Fig. 2b of the paper to make approximate estimates of the model parameters α , β , and ρ . From Fig. 2b, extract the TMG concentrations at the switching threshold and the low- and high-expression levels. Use the saddle-node relations derived in the preceding problem to infer the model parameters.

Finally, compare the inferred values with the corresponding values reported in Fig. 3 and Supplementary Eq. S8, and briefly discuss the limitations of estimating parameters from digitized figures rather than the original experimental data.

Problem 3. A positive-feedback system beyond the lac operon

Choose a biological system other than the lac operon that may contain a positive-feedback loop. Briefly describe the biological mechanism and identify the state variables and their interactions. Draw an interaction diagram that makes the positive-feedback loop explicit.

Search the literature for experimental evidence supporting the proposed positive feedback, and determine whether there is evidence for bistability and/or hysteresis.

Construct a minimal mathematical model of differential equations. Define all the variables and parameters, and explain the biological meaning of each term. Use a production-removal rate plot and phase line to illustrate the fixed points and analyze their stability.

Finally, compare your chosen system with the lac operon system. Discuss the similarities and differences in terms of the source of positive feedback, the nonlinearities or cooperativity, the relevant time scales, and the resulting dynamical behavior, et al.

Problem 4. Damped oscillations in a two-component negative-feedback loop

Consider a two-node negative feedback circuit in which X activates Y according to the increasing function $g(X)$ and Y represses X according to the decreasing function $f(Y)$:

$$\begin{aligned}\frac{dX}{dt} &= \beta_X f(Y) - \alpha_1 X \\ \frac{dY}{dt} &= \beta_Y g(X) - \alpha_2 Y\end{aligned}$$

Here the regulatory input functions f and g are generally nonlinear. The parameters $\alpha_1, \alpha_2 > 0$ are the removal rates of X and Y, while $\beta_X, \beta_Y > 0$ are the maximum production rates.

Let (X^*, Y^*) be the fixed point. Define the local feedback gains at the fixed point as: $\beta_1 = -\beta_X f'(Y^*) > 0$, $\beta_2 = \beta_Y g'(X^*) > 0$.

1. Numerical Analysis: Suppose the input functions are Hill functions of the form

$$\begin{aligned}f(Y) &= \frac{K_Y^n}{K_Y^n + Y^n} \\ g(X) &= \frac{X^m}{K_X^m + X^m}\end{aligned}$$

where $K_X, K_Y > 0$ are the Hill constants and $m, n > 0$ are the Hill coefficients.

Write code to plot the phase portraits for several sets of parameters. For each parameter set,

- plot the vector field and several representative trajectories;
- classify the fixed points as spirals or nodes and analyze their stability.

Bonus: Plot a phase diagram in the (α_1, α_2) parameter space that classifies the different types of fixed points.

2. Prove that sufficiently small perturbations around the fixed point exhibit damped oscillations when

$$(\alpha_1 - \alpha_2)^2 < 4\beta_1\beta_2.$$

Hint: Linearize the nonlinear system around the fixed point. Write down the Jacobian matrix evaluated at the fixed point and solve its characteristic equation. Damped oscillations occur when the eigenvalues have nonzero imaginary parts and negative real parts. Refer to Chapter 5 of Strogatz.

3. **Bonus:** For the nonlinear system above, assume that $f(Y)$ and $g(X)$ are continuously differentiable. Prove that, for any such input functions, the system cannot exhibit sustained oscillations; equivalently, show that there is no limit cycle.

Hint: Calculate the divergence of the vector field and apply the Bendixson–Dulac criterion introduced in Section 7.2 of Strogatz.

Problem 5. Fourier analysis of p53–Mdm2 oscillations

Download the single-cell data of p53 and Mdm2 oscillatory dynamics underlying Figures 1 and 2 of Hsieh et al. (2024) and complete the following tasks.

Paper and data: Hsieh et al., Plausible, robust biological oscillations through allelic buffering, *Cell Systems* 15, 1018–1032.e12 (2024). Single-cell data available at Zenodo: <https://zenodo.org/records/13743894>

1. Plot representative oscillation time courses of p53 and Mdm2 protein levels.
2. Use Fourier analysis to calculate the power spectrum of each p53 and Mdm2 trajectory.
3. For each trajectory, identify the characteristic oscillation frequency and corresponding period.
4. Compare the dominant frequencies of p53 and Mdm2 across cells. Classify the selected cells as displaying approximately 1:1 or 2:1 Mdm2-to-p53 period ratios, and discuss whether the 2:1 relationship is consistent with Mdm2 pulse skipping.
5. (Optional) For cells displaying an approximately 1:1 period ratio, estimate the time lag or phase difference between p53 and Mdm2.

Submit your commented code, figures with clearly labeled axes and units, a table of the estimated frequencies and periods, and a brief interpretation of the results.